

CSE 6242 FINAL POSTER

VISUALIZING AND PREDICTING TRAFFIC CONGESTION RATES

TEAM 3 CUDLY CODERS

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INTRODUCTION

Goal: Forecast and visualize Atlanta traffic congestion using GDOT + weather data.

Motivation:

- Traffic congestion → delays & fuel waste
- Weather affects driving patterns
- Combine both to support data-driven traffic planning

PROBLEM DEFINITION

Task: Time-series prediction of hourly traffic volume for each GDOT site.

Inputs:

- Historical GDOT sensor counts
- Hourly weather (temperature, precipitation, humidity)

Output: Predicted traffic volume (+ congestion level)

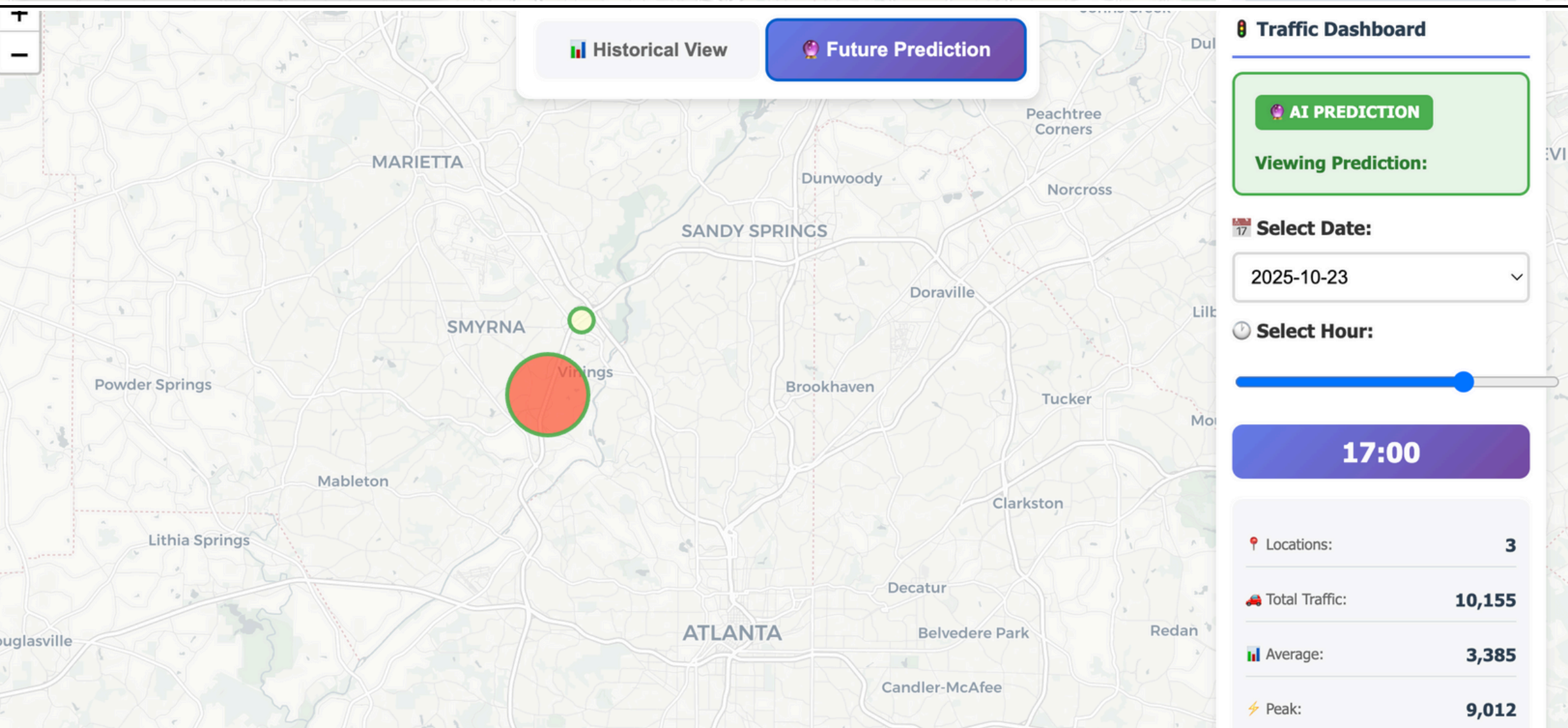
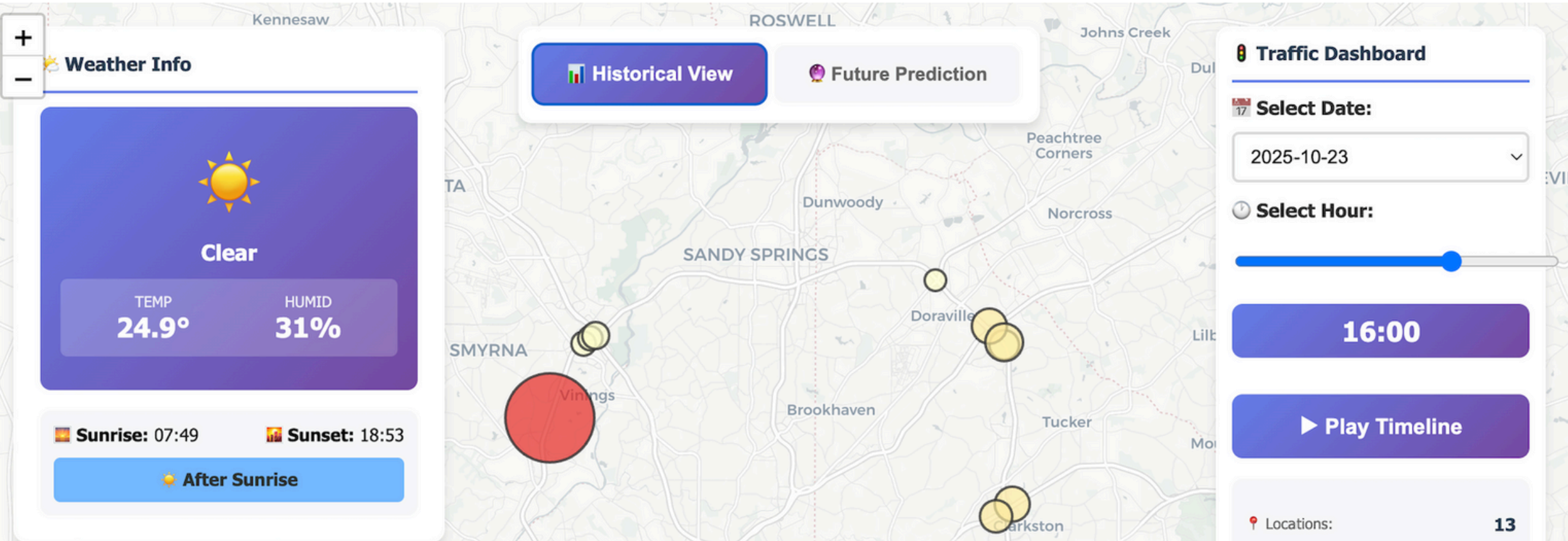
DATA & PREPROCESSING

- Collected GDOT traffic and hourly weather data around I-285.
- Cleaned and merged 400+ Excel files into one dataset (~260K records).
- Aligned timestamps and handled missing values.
- Created time features (hour, day, week) and lag features (1h, 24h, 168h).
- One-hot encoded categorical variables (e.g., wind direction).
- Split data chronologically into train, validation, and test sets.

MAIN PREDICTION MODEL

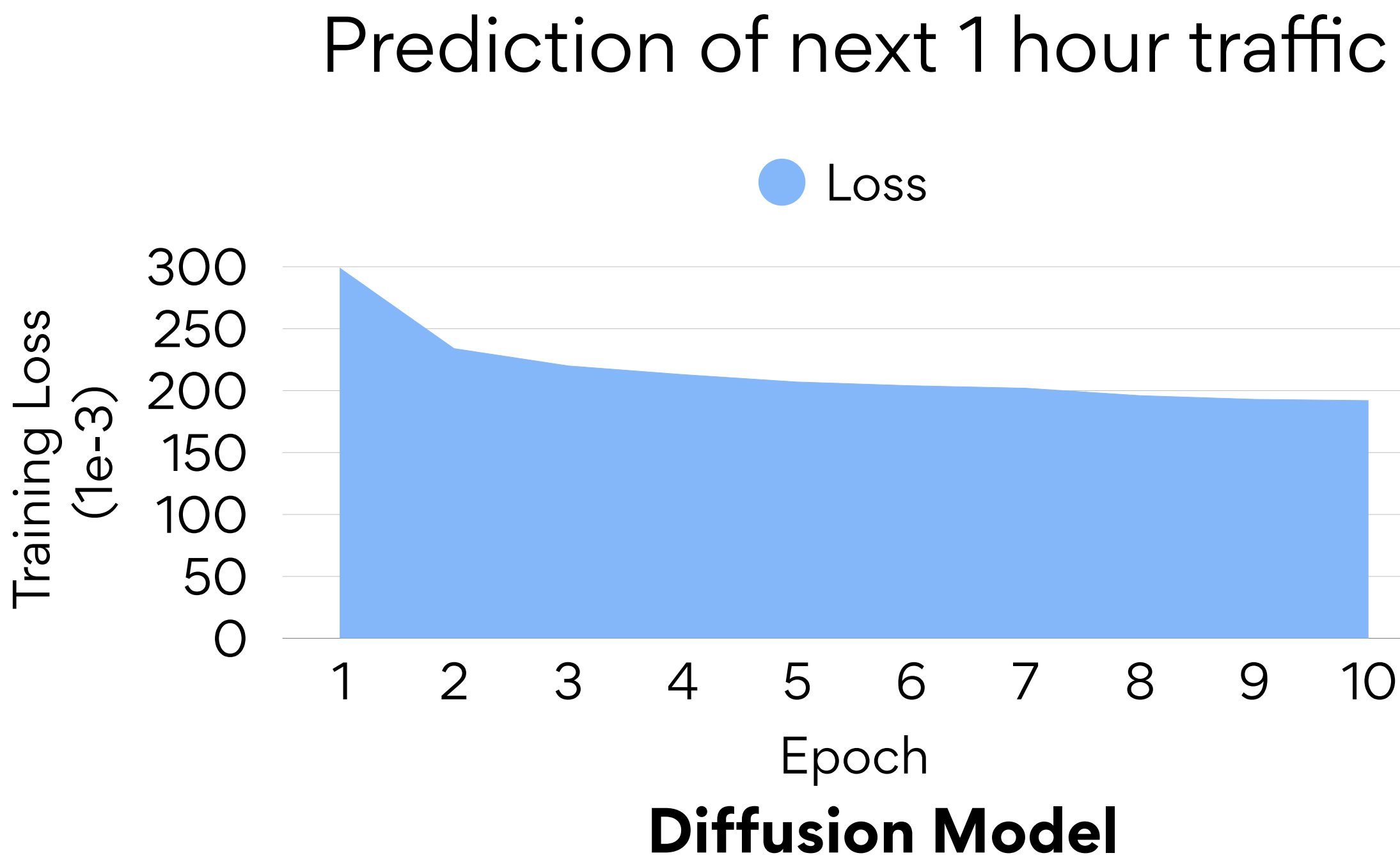
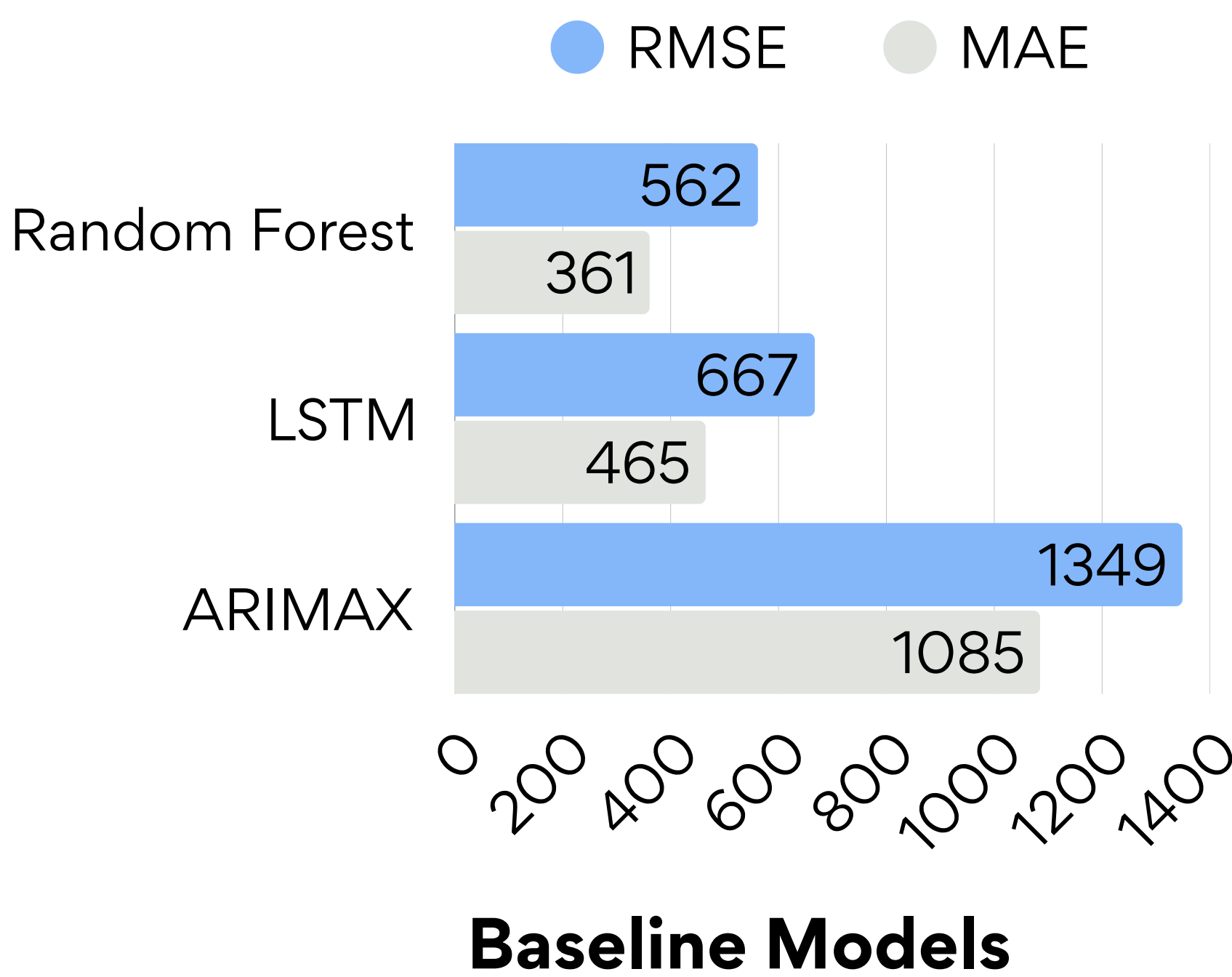
- The DDPM is a generative model that learns to recover clean data from progressively noised samples.
- By conditioning on past traffic and weather context, it generates realistic future traffic volumes through an iterative denoising process.
- Unlike deterministic regressors, DDPM captures the stochastic and multi-modal nature of traffic dynamics, making it well-suited for modeling congestion uncertainty in Atlanta's complex urban network.

VISUALIZATION DASHBOARD



- Developed a web-based traffic visualization dashboard that:
- Displays 330K+ traffic records interactively
- Built using Python + Leaflet.js + HTML/CSS/JS
- Shows traffic intensity (marker size/color = volume)
- Integrates real-time weather overlay
- Provides hour slider + play animation for temporal view
- Compresses data (33MB → 8MB) via gzip for fast client-side loading

BASELINE & DIFFUSION MODEL RESULTS



Test Metrics:

- True Mean: 1539.53
- Mean Squared Error: 260545.05
- Root Mean Squared Error: 510.44
- Mean Absolute Error: 255.27
- R²: 0.95

RESULT & FINDINGS

- Among the baseline models, Random Forest performed best, achieving an RMSE of 562 and MAE of 361. It captured short-term traffic fluctuations more accurately than both LSTM and ARIMAX.
- LSTM learned overall temporal patterns but tended to smooth out sharp changes in traffic volume, resulting in moderate performance.
- ARIMAX showed the weakest accuracy, reflecting its difficulty handling nonlinear and irregular traffic behavior.
- The Diffusion Model produced the strongest overall results, with an RMSE of 510.44, MAE of 255.27, and R² of 0.95. Its training curve shows stable loss reduction, indicating consistent learning throughout training.
- Unlike the baseline models, the diffusion approach models randomness and uncertainty directly, allowing it to better reflect the natural variability in Atlanta's traffic data.
- Overall, the findings show that while baseline models establish useful reference points, the diffusion-based method delivers more accurate and reliable short-term traffic predictions, especially under noisy or rapidly changing conditions.

FURTHER IMPLEMENTATION

- The model currently predicts traffic 1 hour ahead, supporting real-time travel decisions.
- Well-suited for short-term route planning, congestion avoidance, and immediate travel updates.
- The diffusion framework can be extended to 12-24 hour forecasting with additional training.

CONCLUSION

- Diffusion model provides the most accurate and stable short-term predictions compared to baseline models.
- Ideal for real-time use cases such as choosing faster routes or anticipating near-future congestion.
- Baseline models offer useful reference performance but struggle with irregular or noisy traffic patterns.
- Although the current system predicts only 1 hour ahead, results indicate strong potential for multi-hour forecasting.
- Future expansion could enhance both individual traveler guidance and city-level traffic planning.